Science, Universe and Society

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Science, Universe and Society (A29422)

Short Title:	Science, Universe and Society		
Faculty	Science and Computing		
Department	Science		
Cluster	Mathematics and Physics		
Author	CORAIFEARTAIGH		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module allows the student to broaden their awareness of the history of science, of the role of science in society and of the responsibilities of the scientist. In the second part of the module, particular emphasis will be placed on an introduction to the physics of elementary particles. The third part of the module focuses on the science of climate change and the public perception of the problem. The module also continues the development of the professional and independent learning skills of the student.

Programmes

SCIE-0079	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	2 / 3 / M

Indicative Content

- Professional development: reflective practice, presentation skills, working in groups.
- Science and society: social responsibility of scientists, ethical issues in science, public understanding of science, media issues in science.
- Introduction to elementary particle physics: the discovery of atoms, the nuclear atom, the particle zoo and the Standard Model.
- Introduction to climate science: global warming, paleoclimatology and the enhanced greenhouse effect.
- Climate change and the public.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the role of science in society, in particular in relation to ethical and media issues.
2. Describe some introductory components of elementary particle physics.
3. Describe some introductory components of climate science.

Learning and Teaching Methods

- Lectures.
- Tutorials.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	70%	1,2
In-Class Assessment	30%	3

Assessment Criteria

<40%: Inability to demonstrate a basic understanding of the content of the module. Poor writing and presentation skills.

40%-49%: Ability to demonstrate basic professional development skills and an understanding of the role of science in society. Basic knowledge of introductory concepts of frontiers in physics.

50%-59%: Ability to exhibit broadened understanding of the roles of science in society. Good knowledge of introductory concepts of frontiers in physics. Good writing and presentation skills.

60%-69%: Ability to demonstrate very good knowledge of the module content, and present arguments and debate issues relating to this content. Very good writing and presentation skills.

70%-100%: All the previous to an excellent level.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Independent Learning	99	

Essential Material

- Close, F. _Particle Physics: A Very Short Introduction_. Oxford: Oxford University Press, 2004.
- Dessler, A.E. _Introduction to Modern Climate Change_. New York: Cambridge University Press, 2012.

Supplementary Material

- Carey, J. _The Faber Book of Science_. New York: Faber and Faber, 1995.
- Resnik, D.B. _The Ethics of Science_. New York: Routledge, 1998.